

manual mode, if the thumbwheels are set at 2500, then a peak field of 2500 Oe will be reached upon execution of a ramp up or ramp cycle command, if the coil selected is capable of that field. If the coil is not capable of reaching that field, the tracking light will not come on and it is then critical that the field be ramped down manually to avoid coil over heating due to being left on.

## 2.3 Computer Interface

2.3.1 General features. The degaussing system is equipped with a computer interface that enables automatic degaussing under the control of an external computer. The interface is a serial RS232 data link. Connection to the degaussing system is made through a 25-pin D female connector. Signals to the degausser from an external computer are on pin 2; replies from the degausser are on pin 3. Pin 7 is ground. No handshaking protocol is used. The communication baud rate is 1200 and the data word consists of 8 bits with no parity and one stop bit.

Because the data link is implemented in the degausser by a single board computer running a small basic program, the response time of the degausser to commands is slow. For instance, after sending a command DCA1000 (degausser configure amplitude 1000), the degausser will need about one second to process this information before it can accept another command. Suitable wait loops will have to be used in the external computer code to prevent unreliable communications.

2.3.2 Command response protocol. Commands to the degausser and responses all take the form of strings of letters or numbers (ASCII strings). Commands to the degausser must all be preceded by the ASCII letter D. Commands are of three types:

1. Configure
2. Execute ramp
3. Send status

Configure commands set up the system parameters. These commands are denoted by the letter C, e.g., DCCX (Degausser Configure Coil X). Execute ramp commands are denoted by the letter E and cause a ramping of the field up or down or in an up and down (cycle) fashion. The send status command (DSS) causes the degausser to send back its status. A detailed listing of all degausser commands is included at the end of this section.

When the degausser is commanded to send its status, it returns a character string in the following format:

S! R# D# C& A###.#

where

- ! specifies system status and can be Z=zero T=tracking ?=unknown
- # specifies ramp rate parameter and can be any of the following numbers:  
1, 3, 5, 7, 9
- # specifies delay parameter and can be any number between 1 and 9
- & specifies active coil and can be X or Y or Z
- ###.# specifies amplitude and can be any number between 000.0 and 199.9

All commands to the degausser must terminate with a carriage return (ASCII character 13). A line feed should not be sent along with the carriage return. All responses from the degausser are also terminated with a carriage return.

When execute ramp commands are issued, the degausser will automatically answer when the commanded status is received. When a DERU (Degausser Execute Ramp Up) command is given, the system sends back a T when the field is tracking. If tracking does not occur, TRACK ERROR is sent back.

When a DERD (Degausser Execute Ramp Down) command is sent, the system sends back a Z when the field is at zero. If ramp down is unsuccessful, the degausser sends back ZERO ERROR.

When a DERC (Degausser Execute Ramp Cycle) command is given, the degausser sends back a DONE when the cycle is complete. If cycle is unsuccessful, the system sends back TRACK ERROR.

### 2.3.3 Synopsis of degausser protocol 1200 Baud, 8 bits, no parity, one stop bit.

- DCC Degausser Configure Coil (X, Y, Z) default axis Z (see note 1 below)
- DCA Degausser Configure Amplitude (0000-->3000) default amp 0000 (see note 2 below)
- DCD Degausser Configure Delay (1 - 9 seconds) default delay 1 sec.
- DCR Degausser Configure Ramp (1, 3, 5, 7, 9) default 3
  
- DERU Degausser Execute Ramp Up
- DERD Degausser Execute Ramp Down
- DERC Degausser Execute Ramp Cycle

To get status, type:

DSS Degausser Send Status

Reply from controller:

S! R# D# C& A###.#

|                  |                                          |            |           |
|------------------|------------------------------------------|------------|-----------|
| Status(!)        | Z=Zero                                   | T=Tracking | ?=Unknown |
| Ramp (#)         | 1, 3, 5, 7, 9                            |            |           |
| Delay(#)         | 1 to 9 sec: Used only in Ramp Cycle Mode |            |           |
| Coil(&)          | X=X Axis, Y=Y Axis, Z=Z Axis, ?=Unknown  |            |           |
| Amplitude(###.#) | 00 to 3000                               |            |           |

Note 1: When the tracking light is on, the command to configure the coil axis will be ignored. There are two circumstances that cause the tracking light to turn on:

1. Coil is energized and tracking
2. Coil amplitude is set at zero (computer or manual) mode

The second condition is the default condition on power up. To change axis after power up, you must first configure the amplitude to be nonzero.

Note 2: Amplitude commands must contain 4 digits, e.g. if an amplitude of 10 Gauss is required, use the command DCA 0010.

### 3. FIELD CALIBRATION PROCEDURE

The procedures in this section are for both single and multi-axis systems. For single axis systems, ignore references to the X and Z axes.

Field calibration of the degaussing system should be unnecessary if a complete system, including coil and mu metal shields, is delivered. However, if mu metal shields not produced and installed by 2G Enterprises are used, they will usually shift the coil resonate frequency, necessitating field calibration of the system as described below.

1. Initial setup. Set computer/manual switch to manual. Connect all cables between system components (refer to Fig. 2 Degausser System Layout at end of manual for cable interconnections). Remove top cover of Degaussing System Controller.

2. Open loop. This is accomplished by changing the position of the slide switch (S1) mounted near the rear middle (when viewed from the front of the console) of the main degausser control chassis p.c. board. To open loop slide this toggle weitch to the right of the console.

3. Adjust frequency to resonance on X, Y and Z oscillators. Select X axis using front panel switch. Set the two power amplifier front panel attenuation control potentiometers to 1/2 decibels. Use a dual trace oscilloscope; put probe one on either side of R93 (oscillator output) and probe two on the coil current monitor point (test point 1) at the right rear of PC board. Resonance is achieved when the coil current is maximized and is in phase with the oscillator output.

Depress ramp up button. Adjust frequency adjust potentiometer R115X (rear potentiometer on PC board) to maximize the coil current and reduce phase difference between coil current and oscillator output to zero. Push ramp down button. Repeat the above procedure for the Y axis and the Z axis, first selecting the appropriate axis on the front panel switch, then adjusting potentiometers as follows:

| <u>Axis</u> | <u>Oscilloscope</u> |                | <u>Freq. Adjust Pot.</u> |
|-------------|---------------------|----------------|--------------------------|
|             | <u>Probe 1</u>      | <u>Probe 2</u> |                          |
| X           | either side R93     | test point 1   | R115X                    |
| Y           | either side R93     | test point 1   | R115Y                    |
| Z           | either side R93     | test point 1   | R115Z                    |

The X axis frequency adjust potentiometer is located at the right top of the PC board. The Y axis potentiometer is located at the right middle of the PC board and the Z axis potentiometer is near the right front of the PC board.

4. Set up maximum amplifier drive voltage. The purpose of this adjustment is to set the maximum amplifier drive voltage to a value which does not allow output clipping. Open loop as described in paragraph 2.0 above. Set amplifier attenuation potentiometers (front panel) to full on (clockwise). Select the X axis. Put scope ground on the black amplifier terminal of the power amplifier and scope probe on either red terminal of the power amplifier (this allows monitoring of 1/2 of amplifier output). Do not put scope ground probe on the red terminals as this will damage the scope or amplifier. Set scope vertical sensitivity to 50 v/d. This usually requires selecting 10X sensitivity position on the scope probe.

CAUTION: AMPLIFIER VOLTAGE CAN REACH 200 VOLTS PEAK  
ACROSS TWO RED TERMINALS. DO NOT TOUCH!

Push ramp up button. Let the controller ramp up to full voltage (approximately 15-20 seconds). Adjust R134 (near right front of PC board) until amplifier output just barely does not clip at the peak of each cycle. Push ramp down button. Repeat procedure for Y and Z axes, adjusting R135 for the Y axis and R133 for the Z axis. See Figure 7. R133, R134 and R135 are just below left of R115Z.

5. Maximum field calibration. The purpose of this calibration is to enable closed loop tracking up to the maximum field and to calibrate the front panel thumbswitch directly in Gauss. Set the front panel power amplifier attenuation potentiometer to full on and ramp rate to 1 (fastest ramp up). Check to ensure that the loop is open (see paragraph 2 above). FOR MULTIAXIS SYSTEMS, SET THE AXIS SELECT SWITCH TO THE COIL THAT GENERATED THE SMALLEST FIELD AT MAXIMUM AMPLIFIER OUTPUT. For 3 axis nested coil systems, this is the largest diameter coil pair. For 2 axis axial/transverse coil systems, this is the transverse coil pair. For 3 axis identical coil systems, choose the X axis. For single axis systems, choose the X axis (this is normally the axis used for single axis systems).

Set front panel thumbwheel switches at 0000. Put a DVM probe on pin 6 of U66 and adjust R153 (near right center of PC board) to obtain 0.0000 volts.

To perform the next calibration step, it will be necessary to know the coil calibration constant for the coil and the maximum field achievable with the coil. This is normally measured at the factory and supplied with instrument delivery. If you are constructing your own coils, you should measure this using a precision pickup coil of known crosssectional area A and number of turns or another calibrated magnetometer such as a Hall probe. The relationship between the peak applied field  $B_p$  and peak voltage  $V_p$  across this coil is

$$B_p = \frac{V_p}{2\pi fNA} \quad (5.1)$$

where  $f$  is the frequency of the applied (sinusoidal) field. The value of  $V_p$  should be measured with the calibration coil centered in the coil and the amplifier running at maximum output with the loop open.

**CAUTION:** high voltages are present in the degausser coil vicinity. Ensure that adequate insulation (>7000 volts) is present between the calibration coil and the degausser coil. As an added precaution, set up coil and move away from wiring of calibration coil before energizing the degausser. Do not touch calibration coil wiring while degausser is energized. Death can result from shocks from the degausser system.

To provide some leeway for operation when the degausser coils are hot, it is desirable to limit the maximum field actually used in practice to 90% of the ultimate producible field determined above with the loop open. To do this, first determine 90% of the maximum obtainable field and then set the front panel thumbswitches to this value. Ensure that the loop is closed and that the power amplifier is at maximum gain. Put a DVM probe on pin 6 of U66 (near right center of PC board) and adjust R151 (near U66) to obtain a voltage of 5.73 V.

Now insert the calibration coil back into the middle of the degaussing coils (observe precautions noted in the box above!) and arrange a DVM so that the voltage across this coil can be measured. Select AC mode on DVM.

Since the meter reads RMS on AC, determine the desired field level by multiplying the 90% (peak) field value by 0.707. This is the field value that will be tuned to.

Now depress ramp up button (tracking light should come on) and adjust the fine amplitude potentiometer denoted below to obtain the desired field as measured by the calibration coil.

| <u>Axis Selected</u> | <u>Fine Amplitude Potentiometer</u> |                                                                   |
|----------------------|-------------------------------------|-------------------------------------------------------------------|
| X                    | R104                                | (all potentiometers are located<br>near right middle of PC board) |
| Y                    | R136                                |                                                                   |
| Z                    | R103                                |                                                                   |

Having completed amplitude calibration of the coil producing lowest field at maximum amplifier output, it is necessary (in multi-axis systems) to tune the maximum field for the other axes. Only a fine adjustment is required for the additional axes.

Select second axis to be calibrated and insert calibration coil into the degaussing coils for this axis. Thumbwheel switches should remain in the same position as for the previous coil.

Determine DVM reading for this axis for the given maximum (90%) field. This is necessary because the resonate frequency of the second axis may be different from the previous axis. Use formula 5.1 to determine this. Remember to multiply this voltage by 0.707 to determine the RMS voltage corresponding to the new peak voltage. Put DVM probe on calibration coil and ramp up field. Tracking light may or may not come on. Adjust the fine amplitude potentiometer for this axis to obtain correct voltage at calibration coil. Tracking light should be on after this adjustment.

Repeat above procedure for the third axis.

The above procedure sets all axes to have the same maximum field; this field corresponds directly to the front panel thumbswitches.

After finishing the maximum field calibration, close the loop by using the procedures described in paragraph 2 above.

#### 4. FACTORY CALIBRATION PROCEDURES

1. Adjust time delay oscillator frequency. Put counter on pin 3 of U14. Adjust R80 for 10kHz output frequency.

2. Adjust reference output to +10.000 volts. Put 4 1/2 digit DVM probe on pin 6 of U57. Adjust R98 for 10.000 volts output.

3. Adjust window detector for 10 mv width. Put DVM probe on pin 6 of U49. Adjust R100 for 10 mV reading.